BTS-V(S)-04.19-0826 Reg. No. [| | | | | | |]

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B.Tech. Degree V Semester Supplementary Examination April 2019

CS 15-1504 OPERATING SYSTEM

(2015 Scheme)

Time: 3 Hours Maximum Marks: 60

PART A

(Answer *ALL* questions)

(10 × 2 = 20)

I. (a) What are system calls? Explain with an example.

(b) What is meant by race condition? Give an example.

(c) Explain about guaranteed scheduling.

(d) Differentiate between paging and segmentation.

(e) What is meant by inverted page table?

(f) Explain about two phase locking.

(g) What is meant by deadlocks? What are the necessary conditions for a deadlock to occur?

(h) Write notes on clocks and terminals.

(i) Differentiate between security and protection mechanisms.

(j) What are the characteristics of a real time operating system?

PART B

(4 × 10 = 40)

II. (a) What is meant by critical section problem? Explain how semaphores can be used to solve the producer consumer problem. (6)

(b) Briefly explain about inter process communication. (4)

OR

III. (a) What are the different characteristics used to compare scheduling algorithms? (4)

(b) Explain any two CPU scheduling algorithms and compare them for a given example. (6)

IV. (a) Explain any two memory management mechanisms. (4)

(b) With a neat diagram explain about virtual memory address translation based on paging. (6)

OR

V. (a) Write notes on (i) demand paging (ii) associative memory. (4)

(b) Explain about any two page replacement algorithms and compare them. (6)

(P.T.O)

VI. (a) Explain an algorithm for deadlock detection in a system with resources having multiple instances. (6)

(b) Explain about non resource deadlocks and starvation. (4)

OR

VII. (a) What is a resource allocation graph? How is it related with deadlock? (4)

(b) Explain the Banker's algorithm for deadlock avoidance in system with resources having multiple instances. (6)

VIII. (a) Explain about single level, two level and tree structured directory implementation. (5)

(b) Explain about any two disk scheduling algorithms with an example. (5)

OR

IX. (a) Explain in detail about scheduling in RTOS. (5)

(b) Explain about some of the application areas of real time systems (5)

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